90. Diffusion of light in the atmosphere takes place due to

(a) Carbon dioxide
(b) Dust particles
(c) Helium
(d) Water vapour

Ans. (b) : When a ray of light enters a medium where there is a multiplicity of dust and other microscopic particles, the ray of light spreads all around. This is called scattering or diffusion of light. For this reason, the color of the sky appears blue.

91. An oil tanker is partially filled with oil and moves forward on a level road with uniform acceleration. The free surface of oil then

(a) remains horizontal.
(b) is inclined to the horizontal with smaller depth at the rear end.
(c) is inclined to the horizontal with larger depth at the rear end.
(d) Assumes parabolic curve.

Ans. (c) : The surface of tanker will move in forward direction with increasing velocity (as acceleration in uniform), so depth will be larger at the rear end.

92. Which one of the following statements is NOT correct?

(a) The velocity of sound in air increases with the increase of temperature.
(b) The velocity of sound in air is independent of pressure.
(c) The velocity of sound in air decreases as the humidity increases.
(d) The velocity of sound in air is not affected by the change in amplitude and frequency.

Ans. (c) : The speed of sound in air directly proportional to square root of temperature as $C_{air} \propto \sqrt{T}$
Here C_{air} = speed of sound in air
T = temperature of medium
Therefore, as humidity increases speed of sound also increases.
Hence option (c) is not correct statement.

CHEMISTRY

93. With reference to ionic compounds, consider the following statements:

1. Ionic compounds are insoluble in alcohol.
2. Ionic compounds in the solid state are good conductors of electricity.

Which of these statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Ionic compound is a chemical compound in which ions are held together in a lattice structure by ionic bonds. Ionic compounds dissolve in polar solvents, especially those that ionize, such as water and ionic liquids. They are usually appreciably soluble in other polar solvents such as alcohols, acetone. Solid ionic compounds cannot conduct electricity because there are no mobile ions or electrons present in the lattice.

94. Which one of the following statements is correct?

(a) The presence of NaCl increases the rate of setting of Plaster of Paris.
(b) Gypsum is added to the cement to slow down its rate of setting.
(c) All alkaline earth metals from hydrated salts.
(d) Barium and Strontium are found free in nature.

Ans. (d) : Strontium is a relatively abundant element in the Earth's crust. It is as abundant as fluorine and its alkaline earth partner, barium. The most common minerals containing strontium are celestine and strontianite.

95. Regarding the atom of a chemical element, the magnetic quantum number refers to

(a) orientation (b) shape
(c) size (d) spin

Ans. (a) : Magnetic quantum number represents the number of orbitals present in the sub-shell magnetic quantum number about the orientation of the orbital.
An electron due to its orbital motion around the nucleus generates an electric field. This electric field generates magnetic field which in turn interacts with external magnetic field. Thus under the influence of external magnetic field, electrons of a sub shell can orient themselves in a certain preferred region of space around the nucleus called orbital. The magnetic quantum number determines the number of preferred orientation of electron present in a sub shell.

96. Assertion (A) : In the periodic table of chemical elements, electron affinity is always found to increase from top to bottom in a group.
Reason (R) : In a group, the atomic radii generally increase from top to bottom.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (d) : A trend of decreasing electron affinity is found by going down the groups in the periodic table. The additional electron will be entering in an orbital farther away from the nucleus and thus would experience a lesser effective nuclear charge. It is fairly obvious that the atoms get bigger by going down group with the increase in atomic number. The reason is equally obvious, addition of extra layers of electrons.

97. Assertion (A) : Coal-based thermal power stations contribute to acid-rain.
Reason (R) : Oxides of carbon are emitted when coal burns.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (b) : Coal based thermal power plants contribute to acid rain because SO_2 and NO_2 are emitted from these plants which form H_2SO_4 and HNO_3 in atmosphere, that cause acid rain. Oxides of carbon are emitted when coal burns, but it does not contribute to acid rain.

98. Which one of the following statements is correct?

(a) Liquid Sodium is employed as a coolant in nuclear reactors.
(b) Calcium carbonate is an ingredient of tooth paste.
(c) Bordeaux mixture consists of Sodium sulphate and lime.
(d) Zinc amalgams are used as a dental filling.

Ans. (b) : Toothpaste usually contains 20–42% water. The main ingredients of toothpaste are abrasive Chloride and humid. Generally, the amount of abrasive in toothpaste is up to 50%. Ablative contains insoluble particles of aluminum hydroxide (Al (OH) $_3$) and calcium carbonate ($CaCo_3$).

99. Consider the following statements:
The purpose of adding sodium sulphate and sodium silicate to the detergent in a washing powder is
1. to keep the washing powder dry
2. to maintain the alkalinity of the powder
Which of these statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : The addition of silicates to synthetic detergents has proved very beneficial. Silicates soften water by the formation of precipitates that can be easily rinsed away. Soluble silicates contribute to detergents as cleaning aids, processing aids, and corrosion inhibitors. As cleaning aids, soluble silicates provide alkalinity and promote soil suspension. Sodium sulphate is simply a filler in detergent powder, in much the same way as fillers are used in many medical tablets, it doesn't play any active role as such.

BIOLOGY

100. Octopus is

(a) an arthropod (b) an echinoderm
(c) a hemichordate (d) a mollusc

Ans. (d) : Octopus is an animal of class-Cephalopoda and phylum Mollusca. The shell is absent. It is found at bottom of the sea. It kills its prey with poisonous saliva. It can change its colour.

101. Consider the following conditions of a sick human body:
1. Swollen lymph nodes
2. Sweating at night
3. Loss of memory
4. Loss of weight
Which of these are symptoms of AIDS?

(a) 1 and 2 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2, 3 and 4

Ans. (d) : AIDS is a viral disease. In this disease the immune system weakens and the body becomes prone to different type of infection by pathogens. Sweating at night, swollen lymph nodes, loss of memory and loss of weight are also symptoms of AIDS.

102. Consider the following statements:
1. The common blue green algae, Spirogyra and Ulothrix are found in both fresh water ponds and oceans.
2. The chameleon can look ahead with one eye, and at the same time look behind with another.
Which of these statements is/are correct?
Which of these statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : Spirogyra and Ulothrix are green algae not blue algae. The chamelean can see in both the direction forward and backward at the same time.

103. Assertion (A) : Unsaturated fats are more reactive compared with the saturated fats.
Reason (R) : Unsaturated fats have only single bonds in their structure.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (c) : Unsaturated fats are more reactive due to the presence of the multiple bonds between carbon atoms. They undergo a chemical reaction (like hydrolysis, hydrogenation etc.) easily due to the presence of multiple bonds. Hence, the unsaturated fats are more reactive compared with the saturated fats
Unsaturated fats contain one or more double bonds so these molecules are said to be unsaturated.
So, the correct answer is 'Assertion is correct but Reason is incorrect'.